Mobile-Based Virtual Physics Laboratory with AI Assistance and Interactive Learning for Rural Students

R26-IT-150



INTRODUCTION VIDEO



R26-IT-150

PROBLEM

- Physics education requires practical experiments to understandconcepts effectively.
- Many rural schools in Sri Lanka lack laboratory facilities and equipment.
- Students mainly learn physics through theory without hands-onexperience.
- Limited access to teachers and resources reduces understanding ofcomplex concepts.
- Existing learning systems do not integrate experiments, quizzes, andprogress tracking.
- Students do not receive personalized feedback to improve weak areas.

IMPACT

- Poor understanding of physics concepts due to lack of practical experience.
- Students lose interest and motivation in learning physics.
- Lower academic performance and examination results.
- Limited access to experiments decreases students' confidence in science subjects.

PEOPLE WHO BENEFIT FROM THE SOLUTION

- Gain practical physics learning experience through virtual experiments, quizzes, and AI-assisted guidance.
- Can easily demonstrate physics concepts, monitor student progress, and provide better learning support.
- Benefit from a low-cost virtual laboratory solution without needing expensive physical equipment.

DESIGN EXCELLENCE OF THE OVERALL SOLUTION

- Most systems focus only on one feature (experiments, quizzes, or chatbot)separately.
- AI chatbots are not adapted to student grade level or lesson context.
- Learning analytics systems do not combine theory and practical experiment data.
- Many platforms do not provide personalized feedback or learning recommendations.

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Design and Evaluation of the Real World Context-Aware, Grade Adaptive AI Learning Support System for Virtual Physics Laboratories

Specialization :- Information Technology

GAME BASED DYSLEXIA INTERVENTION MODULE

COMMENTS FOR THE PROPOSAL

Comments	Action
Using RAG model only will not be sufficient for a year long project.No functional and non functional requirements.	◦ Extended the component beyond a basic RAG system. ◦ Added Real-World Context-Aware scenario generation using news sources. ◦ Introduced grade-adaptive AI learning support. ◦ Designed a pipeline connecting question and answer models to evaluate answer creativity and relevance. ◦ Improved the research scope with functional and non-functional aspects.

GAME BASED DYSLEXIA INTERVENTION MODULE

How the Solution Addresses the Sub-Problem

- Apply Design Science Research (DSR) to design and evaluate the AI chatbot and workflow system.
- Use physics syllabus content, lessons, and student questions as system inputs.
- Apply LLM and RAG methods for grade-adaptive response generation.
- Implement structured learning workflow for guided navigation.
- Evaluate using accuracy, usability, and student feedback.

User Requirements Addressed

- Easy navigation
- User - friendly interface
- Mobile responsiveness

GAME BASED DYSLEXIA INTERVENTION MODULE

Design Excellence/Contribution

- Build a Grade-Tagged Physics Knowledge Base
- Implement a RAG-Based AI Chatbot Pipeline.
- Develop Grade-Adaptive Response Generation
- Design and Implement the Structured Learning Workflow Module
- Integrate the Chatbot with the Mobile Frontend
- Validate and Evaluate System Performance

User Feedback

- Parents: “Children spend more time learning through interactive physics activities.”
- Students: “The AI chatbot helped me understand difficult physics questions easily.”
- Teachers: “Scenario-based questions improve students’ critical thinking

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Intelligent Keyword-Based Content Retrieval andCurriculum-Aligned 2D Game Generation Module.

Specialization :- Information Technology

Keyword-Based Physics Search and Adaptive Gaming Assistant

COMMENTS FOR THE PROPOSAL

Comments	Action
Presentation skills should be improved.No functional and non functional requirements.	• Focused on physics education for rural students (Grades 9–11) in resource-limited areas. • Intelligent keyword retrieval for notes, formulas, and quizzes mapped to specific grade levels. • A custom framework that scales game difficulty and content based on the student's curriculum progress • Optimized tech stack to ensure high-speed functionality on budget mobile devices.

Keyword-Based Physics Search and Adaptive Gaming Assistant

How the Solution Addresses the Sub-Problem

- Design and evaluate a keyword-based searchand curriculum-aligned gaming system.
- Apply Intelligent Content Retrieval to matchkeywords with Sri Lankan national physics syllabus materials.
- Generate **adaptive 2D educational games** that scope difficulty and content based on the student's grade level.

User Requirements

- User - friendly colorful interface
- Mobile responsive design
- Interactive touch-based games
- Gamified learning experience

Keyword-Based Physics Search and Adaptive Gaming Assistant

Design Excellence

- Implement a keyword-based search engine to retrieve relevant physics explanations and materials.
- Generate 2D educational games specifically aligned with national physics curriculum topics.
- Design a scoping framework to adapt game difficulty and content based on student grade level.
- Map digital learning activities to structured lesson-level objectives for Grades 9 -11.

User Feedback on prototype

- Parents: “Children are motivated to learn physics through educational games and activities.”
- Teachers: “Curriculum-aligned 2D games improve student engagement and concept understanding.”
- Rural Students: “The interactive games make physics practicals easier to understand without a real laboratory”

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Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module

Specialization :- Information Technology

Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module

COMMENTS FOR THE PROPOSAL

Comments	Action
No functional and non functional requirements. Should think about how to carried out the research. No proper understanding about the research.	• Conducted a systematic literature review to identify research gaps in virtual laboratories, adaptive learning, and learning analytics. • Extended the component beyond a basic quiz system by integrating weak student detection with personalized learning paths • Developed AI-driven adaptive quiz and structured question generation based on student performance analysis. • Enhanced the research scope by integrating learning analytics, adaptive feedback, and structured progression within a unified mobile platform.

Intelligent Learning Analytics with Adaptive Feedback-Based Weak Student Remediation and Personalized Instruction Module

How the Solution Addresses the Sub-Problem

- Implemented sub-lesson based quizzes and experiments to solve fragmented assessment of theory and practical learning.
- Introduced adaptive learning paths to address weak student identification and personalized remediation needs.
- Integrated AI-driven feedback and learning analytics to support continuous performance tracking and improvement.

User Requirements

- User - friendly colorful interface
- Mobile responsive design
- Clear feedback messages with visual progress indicators
- Easy navigation between lessons, quizzes, and experiments

Design Excellence

- Develop a secure student login andprofile management system.
- Evaluate theoretical understanding using quizperformance.
- Analyze practical performance using virtualexperiment results.
- Generate learning analytics and personalizedfeedback recommendations.

User Feedback

- Students: “The drag-and-drop experiments make learning physics more enjoyable.”
- Parents: “Children are more interested in learning physics through the mobile application.”
- Teachers: “Interactive simulations help students understand physics concepts more effectively.”

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Interactive Physics Game-Based Measurement, Calculation and Experimentation Learning Module

Specialization :- Information Technology

GAME BASED WORKING MEMORY INTERVENTION MODULE

COMMENTS FOR THE PROPOSAL

Comments	Action
No functional and non functional requirements. Only has virtual experiment implementation using unity.	◦ Clearly defined the scope of the virtual physics practical component for Grade 9,10 and 11 students ◦ Designed interactive virtual laboratory experiments using Unity. ◦ Developed features such as: Drag-and-drop laboratory equipment interaction, Virtual experiment simulations,Experiment timer system

GAME BASED WORKING MEMORY INTERVENTION MODULE

How the Solution Addresses the Sub-Problem

- Develop physics games and virtual experiments.
- Evaluate experiments based on steps, tools, and results.
- Develop physics games and virtualexperiments.
- Capture measurements such as distance, time,and velocity.
- Extract physics measurements from simulationdata.

User Requirements

- User - friendly colorful interface
- Mobile responsive design
- Interactive touch-based games calculations
- Practical learning experience

GAME BASED WORKING MEMORY INTERVENTION MODULE

Design Excellence/Contribution

- Capture measurements .
- Apply physics formulas to generate calculation results.
- Create virtual laboratory experiments with digital tools.
- Evaluate experiment performance.

User Feedback

- Students: “The drag-and-drop experiments make learning physics more enjoyable.”
- Parents: “Children are more interested in learning physics through the mobile application.”
- Teachers: “Interactive simulations help students understand physics concepts more effectively.”

COMMERCIALIZATION/SUSTAINABILITY OF SOLUTION

Customer Persona

- Students (Grade 9,10,11): Primary users who learn physics through virtual experiments, AI chatbot,Quizes and 2D games.

Market Size

- Can be expanded to: Tuition classes , Rural government schools
- Large user base due to increasing demand for digital education and mobile learning applications.

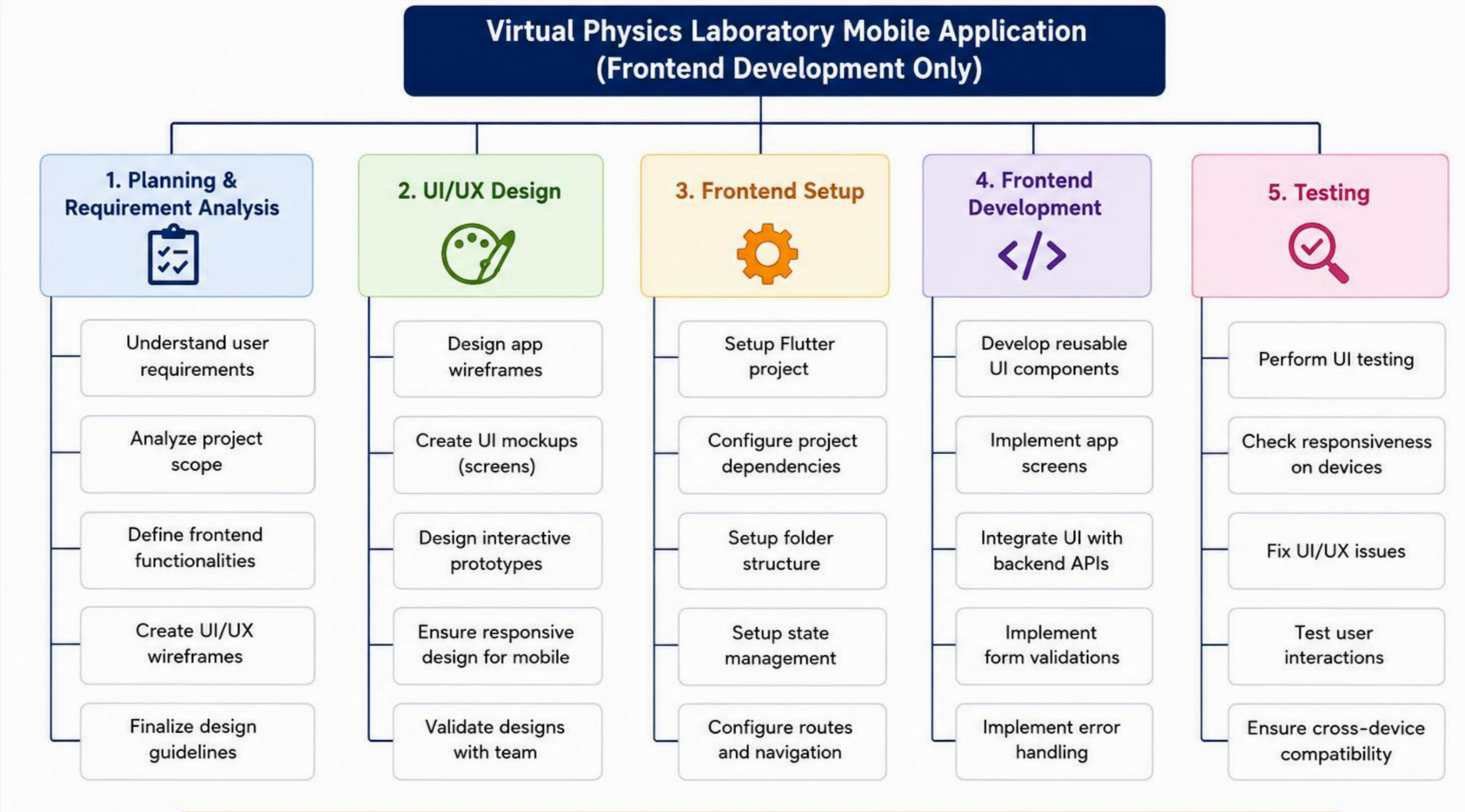
Unique Features

- AI chatbot for physics doubt solving and guidance.
- Keyword-based search engine to find experiments and lessons quickly.
- Virtual physics laboratory
- Adaptive Quiz and performance analysis

How will be the cost covered?

- The system is designed as a low-cost digital solution.
- It can be provided free for students and schools.
- Hosting and data storage managed using Firebase

Work Breakdown Chart



THANK YOU

